

Here is the comprehensive statistical report for `maharashtra_feature_data.csv`:

1. Dataset Scale & Storage

Metric	Value
Total Rows (Observations)	240,679 records
Total Features (Columns)	78 columns
Dataset File Size	176.53 MB (uncompressed) / 41 MB (<code>tar.gz</code>)
In-Memory Footprint	219.25 MB
Completeness	100.0% (0 null values across all 78 columns)
Unique APMC Mandis	367 mandis
Unique Districts / Regions	44 districts / zones
Unique Commodities	129 commodities
Unique Varieties	49 varieties

2. Temporal & Timeline Distribution

- Overall Date Range: January 23, 2024 to September 02, 2026
- Total Calendar Duration: 954 calendar days (~31.3 months / ~2.61 years)

Annual Record Volume			
Year	Record Count	% Share	Active Commodities
2024	54,274	22.55%	10 core crops

2025	155,333	64.54%	127 commodities
2026 (Jan – Sept)	34,287	14.25%	42 commodities
Total	240,679	100.0%	129 commodities

3. Top 15 Commodities by Trading Volume & Price

Rank	Commodity	Record Count	% Share	Mean Modal Price	Median Price	Total Arrivals (Tonnes)
#1	Wheat	38,992	16.20%	₹2,663.1	₹2,600.0	694,324 t
#2	Onion	33,435	13.89%	₹1,920.4	₹1,650.0	12,998,730 t
#3	Tomato	21,191	8.80%	₹1,839.4	₹1,500.0	1,446,289 t
#4	Green Chilli	17,712	7.36%	₹3,925.9	₹3,800.0	263,004 t
#5	Potato	13,909	5.78%	₹1,866.8	₹1,800.0	1,590,466 t
#6	Soyabean	10,971	4.56%	₹4,932.4	₹4,800.0	665,141 t
#7	Bengal Gram (Chana)	8,913	3.70%	₹5,286.7	₹5,250.0	48,574 t
#8	Jowar (Sorghum)	6,105	2.54%	₹2,899.0	₹2,800.0	48,399 t
#9	Rice	5,874	2.44%	₹4,098.0	₹4,000.0	167,713 t
#10	Maize	5,277	2.19%	₹1,798.6	₹1,800.0	353,450 t

#11	Bajra (Pearl Millet)	4,313	1.79%	₹2,246.3	₹2,200.0	187,422 t
#12	Cabbage	3,274	1.36%	₹1,270.8	₹1,200.0	312,851 t
#13	Cauliflower	3,169	1.32%	₹1,614.9	₹1,500.0	291,040 t
#14	Brinjal	3,070	1.28%	₹2,118.2	₹2,000.0	198,477 t
#15	Tur / Arhar (Red Gram)	2,947	1.22%	₹8,420.5	₹8,300.0	42,910 t

4. Crop Group Distribution

Crop Group	Record Count	% Share	Examples
Vegetables	125,664	52.21%	Onion, Tomato, Green Chilli, Potato, Cabbage, Brinjal
Cereals	61,716	25.64%	Wheat, Jowar, Rice, Maize, Bajra
Pulses	18,970	7.88%	Bengal Gram (Chana), Tur (Arhar), Moong, Urd
Oil Seeds	15,108	6.28%	Soyabean, Groundnut, Safflower, Sunflower
Fruits	13,361	5.55%	Banana, Watermelon, Pomegranate, Grapes, Mango
Spices	2,820	1.17%	Chili Red, Turmeric, Garlic, Ginger
Fibre Crops	1,746	0.73%	Cotton, Jute

Others / Specialty	1,294	0.54%	Jaggery (Gur), Flowers, Honey
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5. Price & Arrivals Economic Distribution

Percentile	Modal Price (₹/Quintal)	Min Price (₹/Quintal)	Max Price (₹/Quintal)	Arrivals (Metric Tonnes)
Min	₹100.0	₹50.0	₹150.0	0.05 t (50 kg)
25th (Q1)	₹1,800.0	₹1,200.0	₹2,141.0	1.70 t
50th (Median)	₹2,642.0	₹2,300.0	₹3,000.0	5.40 t
Mean	₹3,415.9	₹2,715.4	₹4,010.6	142.20 t
75th (Q3)	₹4,255.0	₹3,400.0	₹5,000.0	33.00 t
95th	₹7,400.0	₹6,490.0	₹8,500.0	482.00 t
Max	₹90,000.0	₹80,000.0	₹100,000.0	322,600.00 t

6. Geographic Distribution (Top Mandis & Districts)

Top 10 APMC Mandis by Record Count

1. Pune APMC: 7,867 records (3.27%)
2. Nagpur APMC: 6,975 records (2.90%)
3. Pune(Moshi) APMC: 6,102 records (2.54%)
4. Mumbai APMC: 5,976 records (2.48%)
5. Solapur APMC: 5,791 records (2.41%)
6. Pune(Manjri) APMC: 4,765 records (1.98%)
7. Chattrapati Sambhajinagar APMC: 4,742 records (1.97%)
8. Rahata APMC: 4,406 records (1.83%)
9. Satara APMC: 4,164 records (1.73%)
10. Amrawati APMC: 3,855 records (1.60%)

Top 5 Districts	
1.	Pune: 36,089 records (14.99%)
2.	Ahmednagar: 24,416 records (10.14%)
3.	Nashik: 22,790 records (9.47%)
4.	Nagpur: 19,627 records (8.15%)
5.	Jalgaon: 11,905 records (4.95%)

7. Meteorological Weather Profile (Open-Meteo)				
Variable	Column	Min	Mean	Max
Average Temperature	temp_avg	13.8°C	25.0°C	40.6°C
Maximum Temperature	temp_max	21.1°C	30.8°C	47.9°C
Minimum Temperature	temp_min	5.3°C	20.2°C	34.1°C
Daily Precipitation	rainfall	0.0 mm	4.3 mm	253.5 mm
Relative Humidity	humidity	6.1%	67.4%	96.6%
Solar Radiation	solar_radiation	0.8 MJ/m²	17.8 MJ/m²	29.5 MJ/m²
Wind Speed	wind_speed	0.3 km/h	5.8 km/h	41.7 km/h

8. Feature Architecture (78 Columns Breakdown)	
•	16 Core Identifiers & Metadata: date, Commodity, CropGroup, State, Mandi, Variety, Grade, district_name, series_id, plus 7 integer *_id encodings.
•	16 Target & Price Dynamics: ModalPrice, MinPrice, MaxPrice, modal_lag_[1, 3, 7, 14, 30], rolling_mean_[3, 7, 14, 30], rolling_std_[3, 7, 14, 30], price_range, price_spread, norm_spread, volatility_7, 30, zscore_7, momentum_7, 14.
•	8 Arrivals Dynamics: Arrivals, arrivals_lag_[1, 3, 7, 14], arrivals_avg_[7, 30], arrival_change_7.
•	13 Weather & Climate Features: temp_avg, temp_max, temp_min, rainfall, humidity, solar_radiation, wind_speed, temp_range, rain_intensity, temp_avg_30, rain_avg_30, temp_anomaly, rain_anomaly.

- **11 Temporal & Cyclical Features:** `day`, `month`, `year`, `day_of_week`, `week_of_year`, `day_of_year`, `is_weekend`, `sin1`, `cos1`, `sin2`, `cos2`.
- **6 Geolocation & Spatial Features:** `lat`, `lon`, `lat_sin`, `lat_cos`, `lon_sin`, `lon_cos`.